

Curriculum Vitae of Jianyu Hu

Personal Particulars

- Email: jianyu.hu@ntu.edu.sg Tel: (+65) 8441 2344
- Address: Division of Mathematical Sciences, Nanyang Technological University, 21 Nanyang Link, Singapore 637371.
- Homepage: Google Scholar ResearchGate

Research Interests

- Geometry, dynamical systems, statistics, and machine learning, with a particular focus on learning theory for structure-preserving kernel methods.
- Scientific computing, and data-driven learning of dynamical systems.
- Stochastic dynamical systems, operator-theoretic methods, and mathematical physics.
- Stochastic partial differential equations, Onsager–Machlup action functionals, large deviations, most probable transition pathways, and slow manifold reduction.

Education & Research Experience

- Research Fellow, (Advisor: Juan-Pablo Ortega), Dec. 2022 - present, Division of Mathematical Sciences, Nanyang Technological University, Singapore
- Ph.D Candidate, Statistics, (Advisor: Jinqiao Duan), Sep. 2017 - Jun. 2022, School of Mathematics and Statistics, and Center for Mathematical Sciences, Huazhong University of Science and Technology, Wuhan 430074, China
- B.S., Mathematics and Applied Mathematics, Sep. 2012 - Jun. 2016, School of Mathematics and Statistics, Huazhong University of Science and Technology, Wuhan 430074, China

Preprints

1. Kernel learning of PDE solution operators (with JP Ortega). arXiv:2605.09643, 2026.
2. A study of path measures based on second-order Hamilton–Jacobi equations and their applications in stochastic thermodynamics (with Q Huang and Y Huang). arXiv:2508.02469, 2025.
3. A kernel method for the learning of Wasserstein geometric flows (with JP Ortega and D Yin). arXiv:2511.06655, 2025.
4. Learning Hamiltonian systems from trajectories (with DM de Diego, JP Ortega and M Vaquero). Manuscript in preparation, 2026.
5. Learning robotic dynamics on Lie groups (with D Campolo and JP Ortega). Manuscript in preparation, 2026.

Journal papers

1. A global structure-preserving kernel method for the learning of Poisson systems (with JP Ortega and D Yin). *Journal of Nonlinear Science*, 35(4):79, 2025.
2. A Structure-Preserving Kernel Method for Learning Hamiltonian Systems (with JP Ortega and D Yin). *Mathematics of Computation*, 95 (2026), 1719–1774.
3. Data-driven method to learn the most probable transition pathway and stochastic differential equations (with X Chen, J Duan and D Li). *Physica D*, 443:133559, 2023.
4. Nonparametric inference of stochastic differential equations based on the relative entropy rate (with M Dai, J Duan and X Wang). *Mathematical Methods in the Applied Sciences*, 46:2986–2996, 2023.
5. The most likely transition path for a class of distribution-dependent stochastic systems (with W Wei and J Duan). *Stochastics and Dynamics*, 23(08):2340008, 2023.
6. An Onsager–Machlup approach to the most probable transition pathway for a genetic regulatory network (with X Chen and J Duan). *Chaos*, 32(4):041103, 2022.
7. A data-driven approach for discovering the most probable transition pathway for a stochastic carbon cycle system (with J Chen, W Wei and J Duan). *Chaos*, 32(11):113140, 2022.
8. Variational inference of the drift function for stochastic differential equations driven by Lévy processes (with M Dai, J Duan, J Wen and X Wang). *Chaos*, 32(6):061103, 2022.
9. Transition pathways for a class of high dimensional stochastic dynamical systems with Lévy noise (with J Chen). *Chaos*, 31(6):063138, 2021.
10. Most probable transitions from metastable to oscillatory regimes in a carbon cycle system (with W Wei, J Chen and J Duan). *Chaos*, 31(12):121102, 2021.
11. Slow manifolds for dynamical systems with non-Gaussian stable Lévy noise (with S Yuan, X Liu and J Duan). *Analysis and Applications*, 17(03):477–511, 2019.

Conference papers

1. A kernel-based global method for the learning of elastic potentials on Lie groups (with D Campolo, JP Ortega and D Yin). *International Conference on Geometric Science of Information*, 93-102, 2025

Teaching Assistant

- Mathematical analysis, Spring 2018, Huazhong university of Science and Technology, China
- Mathematical analysis, Fall 2018, Huazhong university of Science and Technology, China
- Conducted seminar-style courses on advanced topics in stochastic analysis, stochastic partial differential equations and differential geometry, September 2017 - June 2019, Huazhong university of Science and Technology, China

Invited Talks and Visits

- Visit Prof. Zhou Xiang, Hong Kong University of Science and Technology, Hong Kong, China, February 1–8, 2026.
- AAAI 2026 Workshop, The 40th AAAI Conference on Artificial Intelligence, Singapore, January 20–27, 2026.
- Visit Prof. David Martín de Diego, Instituto de Ciencias Matemáticas (ICMAT), Spain, June 28–July 8, 2025.
- Machine Learning and Dynamical Systems Seminar Series, online, June 5, 2025.
- Joint Doctoral Training Program (funded by China Scholarship Council), University of Illinois at Urbana–Champaign, USA, Sep. 2019 – Sep. 2020.
- Short-term research visit, Illinois Institute of Technology, Chicago, USA, Aug. 2018 – Sep. 2018.

Professional Service

Reviewer for SIAM Journal on Applied Dynamical Systems, Neural Networks, Applied Mathematics Letters, Chaos, and Chaos, Solitons & Fractals.

Conferences

- The 17th World Congress on Computational Mechanics & 10th European Congress on Computational Methods in Applied Sciences and Engineering, Munich, Germany, July 19–24, 2026.
- New Advances in Contact and Symplectic Hamiltonian Dynamics, Madrid, Spain, July 13–17, 2026.
- The 15th AMIS Conference, National and Kapodistrian University of Athens, Greece, July 6–10, 2026.
- Kopernik Forum 2026: Frontiers in Interdisciplinary Mathematics, Shenzhen, China, March 27–28, 2026.
- The 5th Symposium on Machine Learning and Dynamical Systems (MLDS 5) & Differential Equations for Data Science (DEDS 2026), Kyoto University, Japan, February 9–13, 2026.
- GSI'25 – 7th International Conference on Geometric Science of Information, Saint-Malo, France, October 29–31, 2025.
- VII International ICMAT Summer School on Geometry, Dynamics and Field Theory, Madrid, Spain, June 23–27, 2025.
- Workshop on Dynamical Systems and Machine Learning, Tokyo, Japan, February 17–18, 2025.
- Differential Equations for Data Science, Kyoto, Japan, February 11–13, 2025.
- The 10th South Australia Science Conference: "Frontier Interdisciplinary High-level Forum on Applied Mathematics", Shantou, China, November 15–18, 2024.
- International Conference on Scientific Computation and Differential Equations, Singapore, July 14–19, 2024.
- International Conference: Differential Equations for Data Science, online, February 19–21, 2024.
- Non-autonomous Dynamics in Complex Systems: Theory and Applications to Critical Transitions, Max Planck Institute for the Physics of Complex Systems, Dresden, Germany, October 9–27, 2023.
- SIAM Conference on Applications of Dynamical Systems (DS21), online, May 23–27, 2021.
- Summer School of ICTS: Bangalore School on Statistical Physics XIII, online, June 28–July 9, 2021.
- Third Symposium on Stochastic Partial Differential Equations, Huazhong University of Science and Technology, Wuhan, China, June 2021.
- The 6th International Conference on Random Dynamical Systems, Shanghai Normal University, Shanghai, China, July 1–3, 2019.
- Stochastic Partial Differential Equations Conference, Nanjing Normal University, Nanjing, China, August 2018.

- Stochastic Partial Differential Equations and Applications, Henan University, Kaifeng, China, August 2017.
- First Symposium on Stochastic Partial Differential Equations, Wuhan, China, June 2016.